UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

DELIVERABLE No. 4: Systematic Review

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2026 - II

1. Mini Systematic Review

10 studies identified through documented search strings in Scopus and Semantic Scholar

Research question and scope

This mini systematic review examines how recent literature (2022–2026) proposes **reference architectures, enterprise architecture and architectural frameworks** to integrate **generative and agentic AI** (LLMs, agents, multi-agent systems) into **software development** in a governed way, and which **compliance, verification and governance** mechanisms they use. Guiding question:

How should a reference architecture based on specialized generative AI agents be structured to automate and verify the development of critical state information systems, ensuring their compliance with the enterprise architecture and applicable regulations?

The question is aligned with the doctoral problem of governing the use of AI in the development of solutions on top of an institutional reference architecture (application case: the Integrated Pension Information System (SIIP) of the Pension Normalization Office (ONP), whose Reference Architecture requires an Architectural Conformance Assessment (ECA) before development starts and explicitly includes artificial intelligence solutions in its scope).

Search strategy

Database: Scopus (TITLE-ABS-KEY fields). **Time filter:** PUBYEAR > 2021. **Result:** 21 documents. **Execution date:** September 2026.

```
TITLE-ABS-KEY(("generative AI" OR genAI OR "large language model*" OR LLM* OR "agentic AI" OR "AI
agent*" OR "LLM agent*" OR "multi-agent") AND ("reference architecture*" OR "enterprise architecture"
OR "architectural framework*" OR TOGAF OR ArchiMate) AND ("software development" OR "software
engineering" OR "code generation" OR SDLC) AND (complian* OR regulat* OR verif* OR validat* OR
governance)) AND PUBYEAR > 2021
```

The string combines four conceptual blocks (generative/agentic AI · reference/enterprise architecture · software development · compliance/governance) joined with AND. The synonyms inside each block are joined with OR, and wildcards are used to cover word variants.

Database: Semantic Scholar. **Time filter:** publication year 2022–2026 (equivalent to PUBYEAR > 2021 in Scopus). **Result:** 2 documents. **Execution date:** October 2026.

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("generative AI" | genAI | "large language model" | "large language models" | LLM* | "agentic AI" |
"AI agent" | "AI agents" | "LLM agent" | "LLM agents" | "multi-agent") + ("reference architecture" |
"reference architectures" | "enterprise architecture" | "architectural framework" | "architectural
frameworks" | TOGAF | ArchiMate) + ("software development" | "software engineering" | "code
generation" | SDLC) + (complian* | regulat* | verif* | validat* | governance)
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Peruvian databases

ALICIA — CONCYTEC

Filter: (arquitectura de referencia and TOGAF and inteligencia artificial generativa)

Result: zero (0)

RENATI — SUNEDU

Filter: (arquitectura de referencia and TOGAF)

Result: zero (0)

Eligibility criteria and selection procedure

Inclusion	Exclusion
IC1. It addresses generative AI, LLMs or AI agents. IC2. It proposes, evaluates or discusses a reference architecture, enterprise architecture, framework or architectural patterns. IC3. It is set in software engineering / SDLC or in the enterprise integration of these systems. IC4. It deals with compliance, verification, security or governance. IC5. Published between 2022 and 2026 and peer reviewed (article or conference paper).	EC1. Duplicates. EC2. Records without accessible text or without abstract. EC3. Works that use AI only as an analysis tool, with no architectural or governance component. EC4. Language other than English or Spanish. EC5. Grey literature, theses or chapters without peer review.

Procedure.

- Scopus returned 21 documents.
- Semantic Scholar returned 2 documents.
- ALICIA returned 0 (zero) documents.
- RENATI — SUNEDU returned 0 (zero) documents.

The 23 records were screened against IC1–IC5 and EC1–EC5 using title and metadata. The result was 10 included studies (meeting the deliverable's requirement of analyzing 10 papers), 4 records on hold because IC3/IC4 are not evident (abstract pending verification) and 9 excluded. Record-level detail is in Table 1 and the totals are in the PRISMA diagram. The 10 included studies were then read in more detail: in full text (#1, #2, #3 and #8 of Table 2) or from the abstract published by the publisher (#4, #5, #6, #7, #9 and #10).

Search results (n = 23)

Table 1 consolidates the 23 records identified: 21 from Scopus (in the default listing order) and 2 from Semantic Scholar. ALICIA–CONCYTEC and RENATI–SUNEDU returned 0 records. Each record was screened against criteria IC1–IC5 and EC1–EC5 using title and metadata, because the export does not include abstracts. Green: selected; yellow: meets what can be verified but is kept on hold; no color: excluded.

#	Reference (authors, year, title, source)	Database	Type	Cit.	Screening decision
1	Jeong, Cho & Woo (2027*). Architecting Reliable LLM-Integrated Simulation: From Code Generation to Tool-Augmented Interaction in Production Planning. IFIP Advances in Information and Communication Technology, 811.	Scopus	Conference paper	0	Excluded: IC4 not evident.

#	Reference (authors, year, title, source)	Database	Type	Cit.	Screening decision
2	[No authors] (2027+). 25th International Conference on Perspectives in Business Informatics Research, BIR 2026. Lecture Notes in Business Information Processing, 601.	Scopus	Conference proceedings	0	Excluded: EC2 (proceedings without authors or study abstract).
3	Do Val Lopes, Pitta, Belem, Alves & Martins (2026). Engineering AI Agents for Clinical Workflows: A Case Study in Architecture, MLOps, and Governance. Proc. IEEE/ACM 5th Int. Conf. on AI Engineering (CAIN 2026).	Scopus	Conference paper	0	Selected (Table 2, #1).
4	Bandeira, De Oliveira, Maia, Aderaldo & Mendonça (2026). LLM-Native Applications: Toward a New Paradigm in Software Engineering. ICSE Companion 2026.	Scopus	Conference paper	0	Not selected: IC4 not evident in title/metadata (on hold; verify abstract).
5	Alenezi, Akour & Al-Omari (2026). Agentic Artificial Intelligence in Software Engineering: Integration Points, Architectural Frameworks, Empirical Evaluation, and Governance. International Journal of Technology, 17(4).	Scopus	Article	1	Selected (Table 2, #2).
6	Elsisi, Santos & Melo (2026). Lifecycle-Aware GenAI Assistance with MCP via Context Refinement Loops: A Reference Architecture. FSE Companion 2026.	Scopus	Conference paper	0	Not selected: IC4 not evident in title/metadata (on hold; verify abstract).
7	Stein (2026). Toward a taxonomy of generative AI use cases in business contexts: Integrating complexity, risk, and strategy. Journal of Computational Science, 96, 102826.	Scopus	Article	4	Selected (Table 2, #3).
8	Kushnerov, Sokol, Akhiiezer, … Nadtochyi & Balagura (2026). Zero-trust Conceptual Architecture for Secure Integration of Large Language Models and Autonomous AI Agents. ICHORA 2026.	Scopus	Conference paper	2	Selected (Table 2, #4).
9	Aldas-Alarcón & Carrera (2026). Mitigation of Artificial Intelligence Technical Debt and Its Impact on the Success of Institutional Information Systems During the Digital Transformation of Higher Education. ICEDEG (Part F228401).	Scopus	Conference paper	0	Excluded: IC1/IC2 (AI in general, not generative AI/LLM/agents; no explicit architectural component).
10	Siddhu, Pappu & Thanvi (2026). Compliance-Aware Generative AI for Life Insurance Document Synthesis: A Reference Architecture. ICAISET 2026.	Scopus	Conference paper	0	Selected (Table 2, #5).
11	Karuppuchamy (2026). Governed Agentic AI for Software Platforms: A Reference Architecture for Safe Autonomy at Scale. IEEE SoutheastCon 2026.	Scopus	Conference paper	1	Selected (Table 2, #6).
12	Moriarty (2026). From Classroom to Enterprise: Architectural Patterns for LLM Integration. IEEE SysCon 2026.	Scopus	Conference paper	0	Not selected: IC4 not evident in title/metadata (on hold; verify abstract).
13	Enyedi (2026). Human-Certified Module Repositories for the AI Age. IEEE AQTR 2026.	Scopus	Conference paper	1	Selected (Table 2, #7).
14	Pokala (2026). Agentic Dashboard Generation from Natural Language on Lakehouse Platforms. ICAISET 2026.	Scopus	Conference paper	0	Excluded: EC3/IC4 (AI used as a dashboard-generation tool; no governance component).
15	Moraes, Onofre & Aguiar (2026). Safeguarding Big Data Question-Answering Systems with a Two-Layer Input Guardrails. ICEIS 2026, vol. 1.	Scopus	Conference paper	0	Selected (Table 2, #8).
16	Jikeli (2026). Model-Driven Development and Low-Code Platforms in Healthcare Digitalization: Opportunities and Challenges. In Advancements in Digital Health and Care.	Scopus	Book chapter	1	Excluded: EC5 (book chapter); IC1 (does not address generative AI/LLMs/agents).

#	Reference (authors, year, title, source)	Database	Type	Cit.	Screening decision
17	Zeydan & Aydeger (2026). Bias-Free and Auto-Evolving Generative AI: Design Principles, Architectures, and Reinforcement Integration. IEEE CCNC 2026.	Scopus	Conference paper	1	Selected (Table 2, #9).
18	Gadani (2026). Next-Generation Cloud Automation with Generative AI. In Revolutionizing the Cloud: Generative AI, Security and Sustainability.	Scopus	Book chapter	1	Excluded: EC5 (book chapter).
19	Aminiranjbar, Tang, Wang, Pant & Viswanathan (2025). DAWN: Designing Distributed Agents in a Worldwide Network. IEEE Access, 13.	Scopus	Article	7	Not selected: IC3/IC4 not evident in title/metadata (on hold; verify abstract).
20	Joshi (2024). Secure Software Architecture for Enterprise Generative Artificial Intelligence. Int. Scientific and Technical Conf. on Computer Sciences and Information Technologies.	Scopus	Conference paper	2	Selected (Table 2, #10).
21	[No authors] (2024). 14th International Conference on Software Business, ICSOB 2023. Lecture Notes in Business Information Processing, 500.	Scopus	Conference proceedings	—	Excluded: EC2 (proceedings without authors or study abstract); IC1.
22	Vasa (2024). Generative AI and Agentic Orchestration for Data Engineering in Multi-Domain Enterprise Analytics Platforms. International Journal of Artificial Intelligence, Data Science, and Machine Learning, 5(4).	Semantic Scholar	Article	—	Excluded: IC4 not evident (data-quality validation, not AI governance); peer review to be confirmed (EC5).
23	Maire, Badmaev, … & Tirchas (2026). Agentic AI Framework for Technical Excellence: A Scalable Discipline-Based Multimodal Assistant for the Subsurface. IPTC Summit on AI for the Energy Industry.	Semantic Scholar	Conference paper	—	Excluded: IC3/IC4 (subsurface/energy domain; no software engineering or governance focus).

*Table 1. Search results and screening decision. Cit. = Scopus citations (— = no data). Source: own elaboration based on Scopus and Semantic Scholar. * Year assigned by Scopus in the export (later than the search execution date and outside the 2022–2026 range of IC5);*

Selected studies (n = 10)

Table 2 presents the 10 studies that meet IC1–IC5 and do not fall under EC1–EC5. Records #4, #6, #12 and #19 of Table 1 meet the remaining criteria, but IC4 (or IC3) is not evident from title and metadata; they are kept on hold until verified against the abstract. The "Contribution / focus" column indicates the basis of reading: [Full text] when the full article was read, or [Publisher abstract] when only the abstract published on IEEE Xplore was available.

#	Reference	IC1	IC2	IC3	IC4	IC5	Contribution / focus (basis of reading)
1	Do Val Lopes, Pitta, Belem, Alves & Martins (2026). Engineering AI Agents for Clinical Workflows: A Case Study in Architecture, MLOps, and Governance. CAIN 2026. Conference paper.	✓	✓	✓	✓	✓	[Full text] Industry case study of the "Maria" platform (primary healthcare, Brazil, in production), presented as a reference architecture: Clean Architecture plus event-driven architecture, the agent as the unit of modularity with its own MLOps lifecycle, and human-in-the-loop (HITL) governance with an immutable audit record. Observational production evidence (median API latency of 6.13 ms; 0.0% failures on critical agent endpoints; of 144 explicit clinical reviews, 81% approved, 19% corrected and 0% rejected), with no comparative baseline; generalization is left as future work. Regulated domain (healthcare); the AI is the governed system.

#	Reference	IC1	IC2	IC3	IC4	IC5	Contribution / focus (basis of reading)
2	Alenezi, Akour & Al-Omari (2026). Agentic Artificial Intelligence in Software Engineering: Integration Points, Architectural Frameworks, Empirical Evaluation, and Governance. International Journal of Technology, 17(4). Article.	✓	✓	✓	✓	✓	[Full text] Architectural framework (perception, reasoning, memory and action, with governance and observability) and policy-based governance model (policy engine, HITL) for integrating agentic AI into the SDLC. Systematic review (2024–2025) and pre/post quasi-experimental multi-case study of 4 projects in a single corporation (8 baseline weeks and 12 treatment weeks; Wilcoxon test): cycle time reduced by 40–55% and MTTR by 33–48%, with no compliance breaches observed. Stated limits: N = 4, a single organization, no randomization; the requirements and architecture phases are addressed only conceptually.
3	Stein (2026). Toward a taxonomy of generative AI use cases in business contexts: Integrating complexity, risk, and strategy. Journal of Computational Science, 96, 102826. Article.	✓	✓	✓	✓	✓	[Full text] Conceptual taxonomy of generative AI use cases in four complexity levels (A work assistants, B automated code generation, C system-integrated text generation, D tool use) and five perspectives (application context, value creation, strategic alignment, technical autonomy, and data governance/Data Mesh), aimed at enterprise architects. Based on academic and industry literature; the author states that it still requires empirical validation.
4	Kushnerov, Sokol, Akhiiezer, … Nadtochyi & Balagura (2026). Zero-trust Conceptual Architecture for Secure Integration of Large Language Models and Autonomous AI Agents. ICHORA 2026. Conference paper.	✓	✓	✓	✓	✓	[Publisher abstract] Zero Trust architectural framework with mathematical formalization for multi-agent ecosystems (autonomous coding agents): semantic gateway, ephemeral sandboxed environments and continuous authorization using heterogeneous graph neural networks (HGNN). The abstract reports validation by simulation of 100,000 asynchronous transactions, with a true positive rate above 98.7% in detecting remote code execution vulnerabilities and few false positives. Security focus.
5	Siddhu, Pappu & Thanvi (2026). Compliance-Aware Generative AI for Life Insurance Document Synthesis: A Reference Architecture. ICAISET 2026. Conference paper.	✓	✓	✓	✓	✓	[Publisher abstract] Compliance-aware reference architecture that separates probabilistic generation from deterministic compliance verification (retrieval grounding, automated validation and human review) for life insurance documents. Evaluated with 4 LLMs, 50 test cases and 186 comparisons against a baseline: 80% fewer unsupported claims (26.5% to 5.3%) and 100% approval (73% ready to send). Regulated domain (insurance); document synthesis, not software development.
6	Karuppuchamy (2026). Governed Agentic AI for Software Platforms: A Reference Architecture for Safe Autonomy at Scale. IEEE SoutheastCon 2026. Conference paper.	✓	✓	✓	✓	✓	[Publisher abstract] Governance control plane for agentic AI in software platforms, in four layers (LLM-based intent interpretation, policy-aware planning, guarded execution over APIs, and telemetry, auditing and rollback) and “bounded autonomy” (declarative policies, approval gates, runtime validation and allow-lists). Illustrated with enterprise scenarios (AI-assisted feature rollout, incident triage) and prototype-based measurements; the available abstract appears truncated.

#	Reference	IC1	IC2	IC3	IC4	IC5	Contribution / focus (basis of reading)
7	Enyedi (2026). Human-Certified Module Repositories for the AI Age. IEEE AQTR 2026. Conference paper.	✓	✓	✓	✓	✓	[Publisher abstract] Human-Certified Module Repositories (HCMR): reference architecture with a certification and provenance workflow, explicit interface contracts, threat-surface analysis and lessons from recent failures, for the safe assembly of software by humans and AI agents. Discusses governance and accountability; the abstract does not declare an empirical evaluation.
8	Moraes, Onofre & Aguiar (2026). Safeguarding Big Data Question-Answering Systems with a Two-Layer Input Guardrails. ICEIS 2026, vol. 1. Conference paper.	✓	✓	✓	✓	✓	[Full text] Safe-BigQA: security-oriented reference architecture for big data question-answering systems, with two-layer input guardrails (a small language model as a filter and an LLM as a fallback). Jailbreak-detection experiments with 3 datasets and 6 models: F1 of up to 62.87% and a 9.64% improvement over LlamaGuard. Security of a system in operation, not development support; the authors note that effectiveness depends on the quality of the first-stage analysis.
9	Zeydan & Aydeger (2026). Bias-Free and Auto-Evolving Generative AI: Design Principles, Architectures, and Reinforcement Integration. IEEE CCNC 2026. Conference paper.	✓	✓	✓	✓	✓	[Publisher abstract] Framework and reference architecture for bias-free, auto-evolving generative AI pipelines: fairness-aware orchestration, explainability mechanisms, resilience against prompt and data poisoning, and reinforcement learning with a human in the loop, with attention to compliance with AI regulations. The abstract does not declare an empirical evaluation.
10	Joshi (2024). Secure Software Architecture for Enterprise Generative Artificial Intelligence. 2024 IEEE 19th International Conference on Computer Science and Information Technologies (CSIT). Conference paper.	✓	✓	✓	✓	✓	[Publisher abstract] Secure architecture design pattern for generative AI in enterprise applications (security and privacy of sensitive data), illustrated with a technical prototype of a product marketing use case. The abstract does not declare an empirical evaluation; commercial domain.

Table 2. Studies selected according to the inclusion criteria. IC1: generative AI/LLM/agents; IC2: architecture, framework or patterns; IC3: software engineering or enterprise integration; IC4: compliance, verification, security or governance; IC5: 2022–2026, peer reviewed. Source: own elaboration.

Synthesis of findings

• **Architectural proposals predominate.** 9 of 10 studies propose or present an architecture (reference architecture, framework, pattern or reference design): #1, #2, #4, #5, #6, #7, #8, #9 and #10, according to the full text (#1, #2, #8) or the publisher abstract (#4–#7, #9, #10). #3 is a taxonomy of use cases, with no architectural artifact.

• **Governance and compliance are explicit in 7 of 10 studies.** They appear in #1, #2, #3, #5, #6, #7 and #9: agent governance with human-in-the-loop (#1, #6), a framework with policy-based governance and compliance evaluation (#2), risk and data governance (#3), regulatory compliance with deterministic verification separated from generation (#5), certification with provenance and accountability (#7) and compliance with AI regulations (#9). The other three address security: zero-trust (#4), input guardrails (#8) and secure architecture for enterprise AI (#10).

• **Empirical evidence is heterogeneous and generally limited.** 6 of 10 studies report some evaluation: observation in production with no baseline (#1), a pre/post quasi-experiment in 4 projects of a single organization (#2), simulation (#4), a comparison against a baseline over 186 cases (#5), prototype-based measurements (#6, according to the abstract) and jailbreak experiments with 3 datasets (#8). The other 4

declare no empirical evaluation: #3 (it expressly acknowledges this), #7, #9 and #10 (illustrative prototype). Only #2 evaluates agentic AI in real software development projects, with stated limits to generalization.

• **Application contexts.** Enterprise and commercial settings predominate; the explicit regulated domains are healthcare (#1) and insurance (#5), and #2 was applied in a corporation in finance, healthcare and analytics. AI is treated as support for software development or operation in #2 (agentic AI in the SDLC), #4 (autonomous coding agents), #6 (agents in software platforms) and #7 (AI-assisted development); in #1, #5, #8, #9 and #10 the AI is the system to be governed or protected, and #3 classifies use cases (including automated code generation). Neither the full texts (#1, #2, #3, #8) nor the abstracts (#4–#7, #9, #10) refer to digital government, the public sector or social security.

• **Maturity of the field.** 9 of the 10 studies are from 2026 and one from 2024 (#10); there are 8 conference papers and 2 journal articles (#2 and #3); Scopus citations range from 0 to 4. It is an emerging field that is still not consolidated.

Limitations of this review

(i) Database coverage: only Scopus and Semantic Scholar; ALICIA–CONCYTEC and RENATI–SUNEDU returned 0 records. (ii) Screening of the 23 records was done on title and metadata only (the export does not include abstracts) and by a single reviewer; the 4 records on hold (Table 1, #4, #6, #12 and #19), including Elsisi et al. (2026) and Bandeira et al. (2026), which are close to the software life cycle, have not been read and could change the synthesis. (iii) Of the 10 included studies, 4 were read in full text (#1, #2, #3 and #8) and 6 only from the publisher's abstract (#4, #5, #6, #7, #9 and #10; the full text was not available), so for the latter the contribution and empirical validation are those stated in the abstract (that of #6 appears truncated) and the synthesis remains provisional. Data extraction was done by a single reviewer and no formal assessment of methodological quality or risk of bias was carried out. (iv) Grey literature, theses and chapters without peer review were excluded (EC5) and language was limited to English and Spanish (EC4), which leaves out AI governance standards and guidelines (e.g., ISO/IEC 42001, NIST AI RMF). (v) The search was run between September and October 2026.

Access links to the studies

• #1 Do Val Lopes et al. (2026) – Direct access: https://doi.org/10.1145/3793653.3793774; Scopus: https://www.scopus.com/pages/publications/105052340880

• #2 Alenezi et al. (2026) – Direct access: https://ijtech.eng.ui.ac.id/article/view/8380; Scopus: https://www.scopus.com/pages/publications/105046214271

• #3 Stein (2026) – Direct access: https://www.sciencedirect.com/science/article/pii/S187775032600044X; Scopus: https://www.scopus.com/pages/publications/105034631140

• #4 Kushnerov et al. (2026) – Direct access: https://doi.org/10.1109/ICHORA69329.2026.11537001; Scopus: https://www.scopus.com/pages/publications/105042030100

• #5 Siddhu et al. (2026) – Direct access: https://doi.org/10.1109/ICAISET66439.2026.11541931; Scopus: https://www.scopus.com/pages/publications/105042158521

• #6 Karuppuchamy (2026) – Direct access: https://ieeexplore.ieee.org/document/11475963; Scopus: https://www.scopus.com/pages/publications/105037800300

• #7 Enyedi (2026) – Direct access: https://doi.org/10.1109/AQTR70159.2026.11577868 (preprint version: https://arxiv.org/abs/2603.02512); Scopus: https://www.scopus.com/pages/publications/105045435826

• #8 Moraes et al. (2026) – Direct access: https://doi.org/10.5220/0014692300004018; Scopus: https://www.scopus.com/pages/publications/105044791237

• #9 Zeydan & Aydeger (2026) – Direct access: https://doi.org/10.1109/CCNC65079.2026.11366270; Scopus: https://www.scopus.com/pages/publications/105034149112

• #10 Joshi (2024) – Direct access: https://doi.org/10.1109/CSIT65290.2024.10982676; Scopus: https://www.scopus.com/pages/publications/105005829833

• On hold (Table 1, #4) Bandeira et al. (2026) – Direct access: https://doi.org/10.1145/3774748.3795680; Scopus: https://www.scopus.com/pages/publications/105051355847

• On hold (Table 1, #6) Elsisi et al. (2026) – Direct access: https://doi.org/10.1145/3803437.3805617; Scopus: https://www.scopus.com/pages/publications/105045840085

2. PRISMA 2020 Flow Diagram

Diagram built with the official PRISMA 2020 template for new systematic reviews which included searches of databases and registers only (Page et al., 2021), translated into English from the Spanish version. Its phases (identification; screening, which includes reports sought and assessed for eligibility; and inclusion) have non-empty values that are consistent with one another.

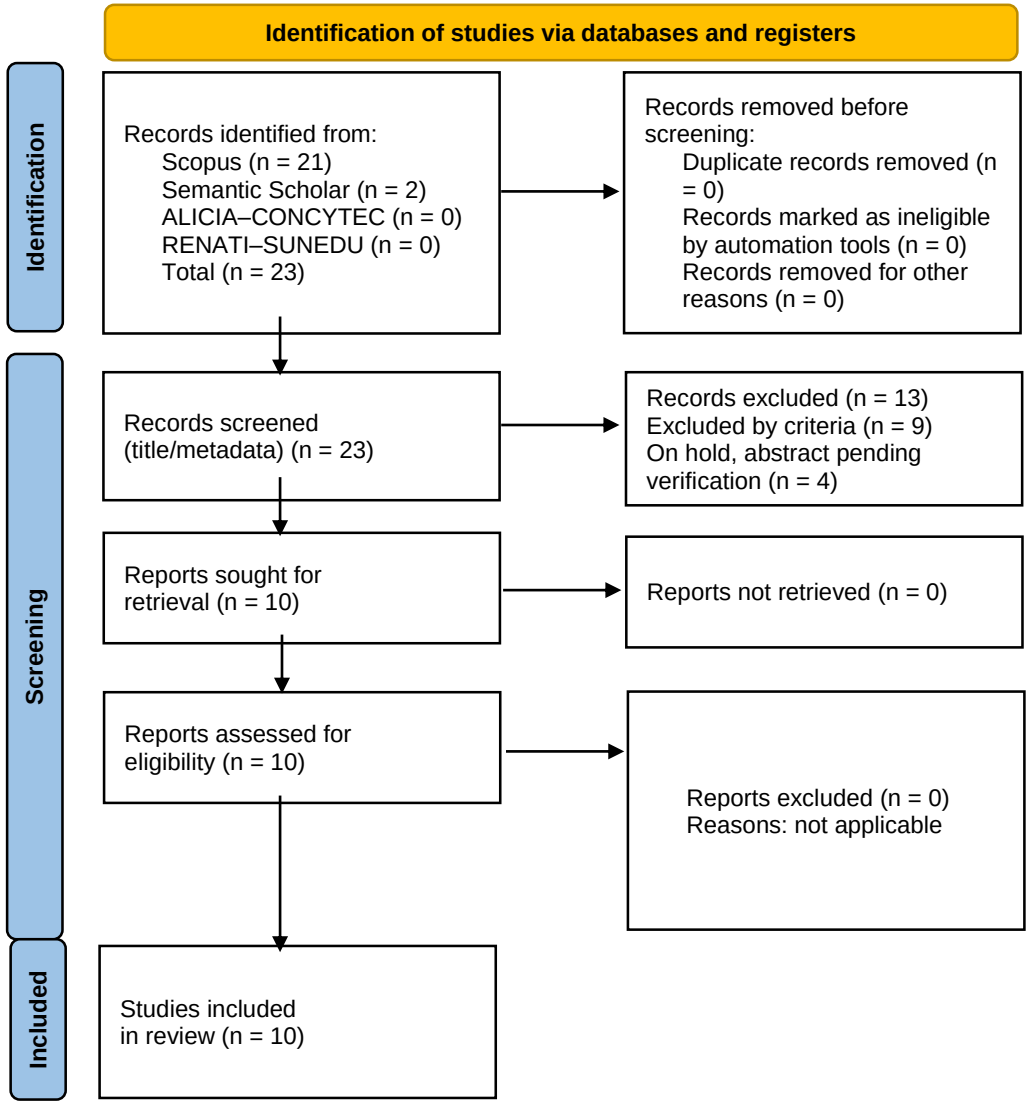


Figure 1. PRISMA 2020 flow diagram of the mini systematic review. Source: own elaboration, adapted from Page et al. (2021).

3. Gap Analysis Table

At least one candidate gap per type (knowledge · methodological · contextual · theoretical), with a citation as evidence.

At least one candidate gap was identified for each type.

Gap type	Candidate gap	Evidence (citation)	Research opportunity
Knowledge	There is no evidence on whether the use of agentic AI in the SDLC remains compliant with formal standards and architectures in contexts other than private enterprise projects.	Alenezi et al. (2026): report that "no compliance breaches were observed" in 4 projects of a single corporation (finance, healthcare and analytics); they warn that the findings are "context-specific and derived from a limited number of enterprise projects" and that the requirements and architectural engineering phases "are mostly conceptual and not extensively addressed".	Study AI compliance in development under a mandatory architectural framework, in a public entity with multiple software factories.
Methodological	Proposals are validated mainly through simulations, test benches or observational cases, and several declare no evaluation; the only pre/post evaluation in real development projects (#2) uses few cases and a quasi-experimental design.	Alenezi et al. (2026): results "indicative rather than broadly generalizable", with a "lack of randomization"; they also point out that enterprise architecture frameworks "are described in terms of conceptualization alone but lack empirical validation within any real-life production environments controlled by governance". Stein (2026): the taxonomy "awaits empirical validation through controlled deployment studies". The abstracts of Enyedi (2026), Zeydan & Aydeger (2026) and Joshi (2024) declare no empirical evaluation; Kushnerov et al. (2026) and Moraes et al. (2026) validate with simulation and test datasets; Siddhu et al. (2026) compare against a baseline on insurance documents.	Design Science Research approach: build a governance artifact and evaluate it in cycles, with compliance metrics (e.g., conformance score) and before/after or control-group comparison.
Contextual	The corpus is concentrated on enterprise and commercial settings; the public sector and social security systems, with legacy system coexistence and State interoperability obligations, do not appear.	Alenezi et al. (2026): a corporation "active in the domains of finance, healthcare, and analytics". Siddhu et al. (2026): life insurance, and Do Val Lopes et al. (2026): healthcare (Brazil), the only explicit regulated domains. Stein (2026): its findings "primarily reflect large enterprise contexts". None of the 10 studies (full texts or abstracts) mentions digital government, the public sector or social security.	SIIP–ONP case: AI in development under the Reference Architecture, with the ECA, legacy coexistence and Peruvian digital government regulations. Verify the gap with the 13 records not included (4 on hold and 9 excluded) and with additional databases.
Theoretical	In the studies reviewed there are autonomy/risk taxonomies and governance proposals, but no theory was identified that links AI autonomy levels to conformance criteria of a reference architecture (TOGAF/ArchiMate).	Stein (2026): A–D taxonomy, aimed at enterprise architects and with technical autonomy and data governance perspectives, which distinguishes layer D as "a fundamental phase change in system autonomy"; it does not link it to conformance criteria of a reference architecture (TOGAF and ArchiMate do not appear in the text). Karuppuchamy (2026) proposes "bounded autonomy" with policies and approval gates, and Kushnerov et al. (2026) continuous zero-trust authorization; their abstracts declare no mapping to architectural conformance.	Theoretical framework that associates each autonomy level (A–D) with controls, artifacts and architectural governance milestones (e.g., the ECA dimensions D1–D5), and subjects it to validation.

References

Alenezi, M., Akour, M., & Al-Omari, O. (2026). Agentic artificial intelligence in software engineering: Integration points, architectural frameworks, empirical evaluation, and governance. International Journal of Technology, 17(4), 1245–1262. https://ijtech.eng.ui.ac.id/article/view/8380

Bandeira, A. P., De Oliveira, G. L. B., Maia, P. H. M., Aderaldo, C., & Mendonça, N. (2026). LLM-native applications: Toward a new paradigm in software engineering. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering: Companion Proceedings (ICSE Companion 2026). https://doi.org/10.1145/3774748.3795680

Do Val Lopes, C. L., Pitta, J. M., Belem, F. M., Alves, G., & Martins, F. V. C. (2026). Engineering AI agents for clinical workflows: A case study in architecture, MLOps, and governance. In 2026 IEEE/ACM 5th International Conference on AI Engineering – Software Engineering for AI (CAIN '26). https://doi.org/10.1145/3793653.3793774

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